

Case Study: How St James's Hospital's Journey to Cloud Transformed Cancer Care

Client: St James's Hospital, Dublin

Alliance: EY and Microsoft

Industry: Healthcare

Focus: Cloud Transformation, Data Modernization

1. Objective

To help St James's Hospital modernize its legacy infrastructure through cloud adoption, improving diagnostics, compliance, and patient outcomes in cancer care.

2. Background

St James's Hospital is one of Ireland's largest cancer centers and an academic teaching hospital affiliated with Trinity College Dublin.

It's known for delivering high-quality care and for its advanced laboratory services, including next-generation sequencing and molecular genetic testing - both essential in diagnosing and researching cancer.

Over time, the hospital's aging hardware created several challenges:

- Slow data processing and delayed diagnostics
- Difficulty maintaining ISO 15189 compliance
- Limited capacity to store large genomic datasets
- Frustrating user experience for laboratory teams

These limitations made it clear that a move to a modern, cloud-based infrastructure was critical.

3. Problem Statement

How can modernizing legacy systems through cloud technology enhance diagnostic efficiency, strengthen compliance, and deliver faster, more accurate patient outcomes?

4. Approach

EY partnered with Microsoft and the Health Service Executive (HSE) to guide St James's Hospital through its digital transformation. The goal was to build a secure, scalable, and compliant cloud solution using Microsoft Azure.

Key steps taken:

- Migrated on-premise data systems to Azure Cloud
- Designed a flexible and secure data architecture
- Used Microsoft Fabric for parallel data analysis pipelines
- Introduced a shared remote desktop for 24/7 access to sequencing tools
- Strengthened data protection and recovery systems

This approach ensured the hospital could process and manage sensitive genomic data efficiently while maintaining compliance and operational continuity.

5. Tools and Technologies

- **Cloud Platform:** Microsoft Azure
 - **Data Tools:** Microsoft Fabric, Advanced Analytics, Machine Learning
 - **Compliance Framework:** ISO 15189 (Medical Laboratories Standard)
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6. Results and Impact

The transformation brought measurable improvements:

- Faster, more secure data management through Azure
 - Enhanced scalability for large genomic workloads
 - Laboratory processing time reduced from **nine hours to under two hours**
 - **50% reduction** in report turnaround time
 - Increased capacity to analyze over **300 genetic markers** (previously only 30)
 - Improved collaboration and efficiency across scientific teams
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7. Key Outcomes

- **Compliance:** Fully aligned with ISO 15189 standards
 - **Efficiency:** Automated overnight analysis and real-time access
 - **Data Accessibility:** Unified systems for smoother collaboration
 - **Innovation:** Leveraged analytics and machine learning for deeper insights
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8. Collaboration Highlights

The project was a joint effort between:

- St James's Hospital scientific teams
- EY Cloud and Health Transformation group
- Microsoft Ireland Health Division
- Health Service Executive (HSE)

This close collaboration fostered innovation, trust, and shared learning, making it a model for future healthcare modernization projects.

9. Voices from the Team

“The most important achievement from this project has been faster processing times. We’ve been able to run more samples and tests, ensuring that each patient gets the best diagnosis for them.”

- *Fiona Kearns, Lab Manager, St James’s Hospital*

“This transformation showcases the power of collaboration and innovation - revolutionizing diagnostics, resolving compliance challenges, and empowering a digital workforce.”

- *Paul Browne, Head of Microsoft and Cloud Services, EY Ireland*

10. Conclusion

St James’s Hospital’s move to the cloud marks a turning point in modern healthcare. By embracing cloud technology, the hospital now delivers faster, more accurate genetic testing, stronger data protection, and better patient outcomes.

This initiative shows how technology, when guided by collaboration and purpose, can truly transform care delivery - setting a new standard for hospitals across Ireland and beyond.

11. Reference

[How St James’s Hospital’s Journey to Cloud Transformed Cancer Care – EY](#)